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Ordinary Differential Equation Notes

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July 15, 2026

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1 First Order Differential Equations

For the first order differential equations, we would consider to solve

$$\frac{\mathrm{d}y}{\mathrm{d}t} = f(t, y)$$

Consider a simpler case:

$$\frac{\mathrm{d}}{\mathrm{d}} = g(t)$$

One with elementary calculus knowledge can solve it:

$$y(t) = \int g(t)dt + c$$

Consider other simpler cases, as f is consisted by linear equations.

Definition 1.1 (First-oder linear differential equation)**.**

The general first-order linear differential equation is

$$\frac{\mathrm{d}y}{\mathrm{d}t} + a(t)y = b(t) \tag{1}$$

Needless to say, $a(t)$, $b(t)$ are assumed to be continuous functions of time. To slove general first order linear equation(1), we could consider a simpler case:

Definition 1.2 (Homogeneous first order linear equation)**.**
The equation

$$\frac{\mathrm{d}y}{\mathrm{d}t} + a(t)y = 0 \tag{2}$$

is called homogeneous first-order linear differential equation

Now we try to solve them part by part.

1.1 Homogeneous first order linear differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}t} + a(t)y = 0.$$
$$\frac{\mathrm{d}y}{\mathrm{d}t} = -a(t)y.$$
$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}t} = -a(t).$$

Integrating both sides with respect to t gives

$$\int \frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}t}\,dt = -\int a(t)\,dt.$$

By the chain rule, $\frac{1}{y}\frac{dy}{dt} = \frac{d}{dt}\ln|y|$, hence

$$\ln|y(t)| = -\int a(t)\,dt + C,$$

where C is an arbitrary constant of integration. Exponentiating both sides yields

$$|y(t)| = e^{C}e^{-\int a(t)\,dt}.$$

Since $e^C > 0$, we may absorb the sign into a new constant $c = \pm e^C$, obtaining the general solution

$$\boxed{y(t) = c\,e^{-\int a(t)\,dt}} \tag{3}$$

where $c \in \mathbb{R}$ is an arbitrary constant. Note that the zero solution corresponds to $c = 0$.

In most cases, we are not interest in all solutions of (3). We could add more assmuption as

$$y(t_0) = y_0 \tag{4}$$

Then we integral (3) between t_0 and t.
Recalling $\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}t} = -a(t)$, we could write:

$$\int_{t_0}^{t} \frac{\mathrm{d}}{\mathrm{d}s}\ln|y(s)|ds = \int_{t_0}^{t} -a(s)ds$$

So

$$\ln|y(t)| - \ln|y(t_0)| = \ln|\frac{y(t)}{y(t_0)}| = -\int_{t_0}^{t} a(s)ds$$

Thus,

$$|\frac{y(t)}{y(t_0}| = \exp(-\int_{t_0}^{t} a(s)ds)$$

The problem now becomes +1 or -1. By taking $t = t_0$, we can find it's +1.
So

$$y(t) = y(t_0)\exp(\int_{t_0}^{t} a(s)ds) \tag{5}$$

1.2 General first order linear differential equation

Now consider the general (inhomogeneous) first-order linear equation:

$$\frac{\mathrm{d}y}{\mathrm{d}t} + a(t)y = b(t). \tag{GL}$$

We solve it using the *integrating factor* method. The key idea is to multiply both sides of (GL) by a function $\mu(t)$ (the integrating factor) chosen so that the left-hand side becomes an exact derivative.
Let $\mu(t)$ be a differentiable function to be determined. Multiplying (GL) by $\mu(t)$ gives

$$\mu(t)\frac{\mathrm{d}y}{\mathrm{d}t} + \mu(t)a(t)y = \mu(t)b(t). \tag{1}$$

We wish the left-hand side to equal $\frac{d}{dt}\big(\mu(t)y(t)\big)$. By the product rule,

$$\frac{d}{dt}\big(\mu y\big) = \mu\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{\mathrm{d}\mu}{\mathrm{d}t}y.$$

Comparing with (1), we require

$$\frac{\mathrm{d}\mu}{\mathrm{d}t} = \mu(t)a(t).$$

This is itself a separable first-order ODE for μ:

$$\frac{1}{\mu}\frac{\mathrm{d}\mu}{\mathrm{d}t} = a(t).$$

Integrating,

$$\ln|\mu(t)| = \int a(t)\,dt \quad \Longrightarrow \quad \mu(t) = e^{\int a(t)\,dt},$$

where we take the principal (positive) branch since only one integrating factor is needed.
With $\mu(t)$ so defined, equation (1) becomes

$$\frac{d}{dt}\big(\mu(t)y(t)\big) = \mu(t)b(t).$$

Integrating both sides with respect to t,

$$\mu(t)y(t) = \int \mu(t)b(t)\,dt + C,$$

where C is an arbitrary constant. Solving for $y(t)$ yields the general solution

$$\boxed{y(t) = \frac{1}{\mu(t)}\left(\int \mu(t)b(t)\,dt + C\right)}, \qquad \mu(t) = e^{\int a(t)\,dt}.$$

Verification. Differentiate the candidate solution:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = -\frac{\frac{\mathrm{d}\mu}{\mathrm{d}t}}{\mu^2}\left(\int \mu b\, dt + C\right) + \frac{1}{\mu}\cdot \mu b = -a(t)y + b(t),$$

where we used $\frac{\mathrm{d}\mu}{\mathrm{d}t} = \mu a$. Hence $\frac{\mathrm{d}y}{\mathrm{d}t} + a(t)y = b(t)$, confirming the solution.

Example. Solve $\frac{\mathrm{d}y}{\mathrm{d}t} + 2ty = t$.
Here $a(t) = 2t$, $b(t) = t$. The integrating factor is

$$\mu(t) = e^{\int 2t\, dt} = e^{t^2}.$$

Then

$$y(t) = e^{-t^2}\left(\int e^{t^2}\cdot t\, dt + C\right).$$

Let $u = t^2$, $du = 2t\, dt$, so $\int te^{t^2} dt = \frac{1}{2}\int e^u du = \frac{1}{2}e^{t^2}$. Hence

$$y(t) = e^{-t^2}\left(\frac{1}{2}e^{t^2} + C\right) = \frac{1}{2} + Ce^{-t^2}.$$